MINISTRY OF SCIENCE & TECHNOLOGY

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INTER - UNIVERSITY STARTUP CHALLENGE

NATIONAL SCIENCE FOUNDATION

TEAM PROPOSAL

TEAM NAME	Axiom V
UNIVERSITY	Informatics Institute of Technology
PRODUCT NAME	Skill.IQ
CATEGORY	EdTech / HRTech (AI-Powered Career Readiness & Recruitment)

Organized by the Industrial Management Science Students' Association,
Department of Industrial Management,
Faculty of Science,
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1. Executive Summary

Skill.IQ is a dual-sided AI platform built to close the widening gap between what universities teach and what employers actually need. On one side, graduates and career-switchers arrive job-capable but interview-anxious; on the other, companies spend an average of 23 hours screening resumes per hire and often wait 40+ days to fill a single role, largely because existing tools solve only half the problem, platforms like Coursera build skills without validating readiness, while LinkedIn lists jobs without ever proving a candidate can perform in one.

Skill.IQ closes that loop. Candidates describe their career goals through a conversational AI interface, which builds a personalized, industry-approved learning path, verifies progress through checkpoints, and prepares them through Smart AI Mock Interviews that assess both technical answers and behavioral confidence using sentiment analysis. Once a candidate passes, they complete job-simulation assignments before Skill.IQ proactively recommends them to companies, flipping recruitment from a search process into an invitation process.

For companies, this means no more posting-and-praying: they describe the role they need filled, and Skill.IQ's AI delivers a shortlist of pre-vetted candidates, each backed by a Performance Report that replaces guesswork with evidence. The result is a platform where demonstrated skill, not a polished resume, is what gets a candidate hired.

2. Problem Definition & Solution

2.1 Problem Definition

The core problem Skill.IQ addresses is the Education-Employment Gap: graduates and early-career professionals complete their education without industry-ready skills or interview confidence, leaving thousands unemployed or underemployed despite holding degrees. On the other side of the same gap, companies, particularly tech SMEs without large in-house recruiting teams, face high Time-to-Hire metrics and inefficient, manual screening processes that cost them time, money, and often the right candidate. This problem affects two distinct but interdependent groups: job-seeking graduates who lack a bridge into the workforce, and hiring managers who lack a reliable way to identify who is actually ready to perform.

2.2 Problem Analysis

The scale of this gap is well documented. According to ManpowerGroup's global Talent Shortage research, roughly 75% of companies report difficulty finding talent with the right mix of technical and soft skills, despite record numbers of graduates entering the workforce each year. On the hiring side, HR managers spend an average of 23 hours screening resumes for a single role, yet a large share of shortlisted candidates still fail at the first interview stage due to a lack of preparation and confidence, not a lack of capability.

Existing solutions each solve only one side of this problem. Learning platforms such as Coursera and Udemy provide content and certification, but stop at "course completed", they offer no mechanism to prove a learner can apply that skill under real interview or job conditions, and no connection to employers. Job platforms such as LinkedIn and Indeed operate on a "Post-and-Pray" model: they generate volume and noise, but provide employers with no validation of a candidate's actual readiness, forcing HR teams to manually filter out unqualified applicants at scale. A newer category of AI interview-prep tools, such as Yoodli and Big Interview, has emerged to help candidates rehearse, but these tools are also one-sided: they improve a candidate's delivery in isolation and have no pathway to actually connect that improved candidate with a hiring company. Similarly, employer-facing AI interview tools like HireVue automate screening for large enterprises but do nothing to prepare or upskill the candidate beforehand, and are priced and built for enterprise ATS integration rather than early-career, SME-scale hiring.

None of these existing players close the loop from "unprepared graduate" to "verified, hired employee", which is precisely the gap Skill.IQ is built to fill.

2.3 Proposed Solution

Skill.IQ is a dual-sided ecosystem that treats career readiness and recruitment as one continuous process rather than two separate markets. On the candidate side, an AI career coach assesses a user's goals, builds a personalized and industry-approved learning path, verifies genuine progress through checkpoints and tests, and prepares the candidate through repeatable AI-driven mock interviews that score both technical substance and behavioral delivery. On the company side, the same underlying AI acts as a proactive recruiter: businesses describe the role and expectations they need filled, and Skill.IQ's engine surfaces only candidates who have already proven, through the platform itself that they are ready to succeed in that role, each backed by a data-driven Performance Report. By making demonstrated skill the trigger for a match, Skill.IQ removes the inefficiency, guesswork, and anxiety from both sides of hiring.

3. Product Overview

3.1 Product Overview

Skill.IQ is an AI-powered career coach and proactive recruiter in one product. On the candidate side, the platform opens with a conversational assessment: the AI asks the user about their career goals and background, then follows up with clarifying questions to build a genuine picture of their aspirations and current skill level. From there, it constructs a structured, industry-approved learning path and enforces accountability through periodic checkpoints, tests, and submissions, ensuring the learner actually absorbs the material rather than passively consuming it. Once the learning path is complete, the AI schedules Smart Mock Interviews, using an anti-cheating verification system to ensure authentic responses, and continues coaching and re-testing the candidate until they demonstrably pass and users who didn’t need the learning paths will directly move into this phase. Successful candidates then move into job-simulation assignments that mirror real on-the-job tasks, building practical experience before the AI proactively matches them with companies and facilitates direct contact for interviews, continuing the search on the candidate's behalf until they are hired.

On the company side, businesses access a dedicated chat interface to describe the employees they need and what they expect from them. The AI conducts a deeper examination of those requirements to understand the role, skillset, and cultural fit being sought, then surfaces pre-qualified candidates from the platform who are already interviewed and validated for that specific type of role along with their CVs. This removes advertising costs, cuts the waiting period for applications, and gives companies access to high-quality candidates who arrive with demonstrated, platform-verified readiness.

The long-term vision is to make "applying for jobs," in its current form obsolete, replacing it with a system where a candidate's demonstrated skill is itself the application.

3.2 In-Scope & Out-of-Scope

In-Scope: AI-driven learning path construction and progress tracking, voice- and text-based AI mock interviews with behavioral and sentiment analysis, an automated candidate-to-company matching and notification system, and a company-facing dashboard for describing hiring needs and receiving pre-vetted candidates.

Out-of-Scope: Skill.IQ is not a content creator, it aggregates and curates existing third-party courses rather than producing original curriculum. It does not handle legal employment contracts, offer letters, or payroll processing, positioning itself strictly as the readiness and matching layer between education and employment, not as an employer-of-record or HR compliance platform.

3.3 Uniqueness & Special Features

Skill.IQ's differentiation rests on four mechanics that existing platforms don't combine:

- Proactive Matching : instead of candidates searching and applying, the AI actively recommends and "sells" a successful candidate directly to a company, flipping the traditional job-search model on its head.

- Independent Validation : every candidate is issued an AI-generated Performance Report from their mock interview performance, giving employers a level of confidence no resume can offer on its own.

- Locked-Progression Accountability : career-matching features remain locked until a candidate's learning path is fully verified, guaranteeing that every candidate surfaced to a company has met a consistent quality bar.

- Cultural Alignment Simulation : companies can customize the AI's interview style to reflect their own internal interview process, so candidates are assessed for cultural fit alongside technical skill.

4. Market Analysis & Business Strategy

4.1 Target Audience & Market Opportunity

Skill.IQ serves two distinct customer segments. Users are primarily Gen Z students and early-career professionals, as well as career-switchers seeking a structured, validated path into a new industry, a segment operating inside a global EdTech market valued at USD 254.8 billion in 2021 and projected to reach USD 605.4 billion by 2027, growing at a CAGR of roughly 15.5%. Customers are tech SMEs and HR departments that are frequently priced out of premium recruitment agencies (which typically charge 20–25% of a hire's first-year salary) and are overwhelmed by manual, resume-based screening, making them a strong early-adopter segment for a lower-cost, pre-vetted matching service.

4.2 Competitor Analysis

- LinkedIn / Indeed (Job Boards): Large networks and high reach, but operate on a "Post-and-Pray" model — high applicant noise and no validation of a candidate's actual readiness. Skill.IQ offers a smaller, but pre-interviewed and quality-assured, candidate pool.

- Coursera / Udemy (Learning Platforms): Strong content libraries, but stop at course completion with no job-placement mechanism and no proof of applied skill. Skill.IQ integrates the learning itself into a verified, employment-linked pathway.

- Yoodli / Big Interview (AI Interview Coaching): Useful for improving a candidate's delivery and communication in isolation, but disconnected from any actual hiring pipeline — practice never converts into an employer introduction. Skill.IQ closes that gap by connecting interview-readiness directly to real company matches.

- HireVue (Employer-Side AI Screening): Strong enterprise ATS integration for automating a company's own interview screening, but does nothing to prepare or upskill the candidate beforehand, and is built for large enterprise hiring volume rather than SME-scale, cost-sensitive hiring. Skill.IQ serves the segment HireVue doesn't: growth-stage companies that need pre-vetted candidates without enterprise pricing or integration overhead.

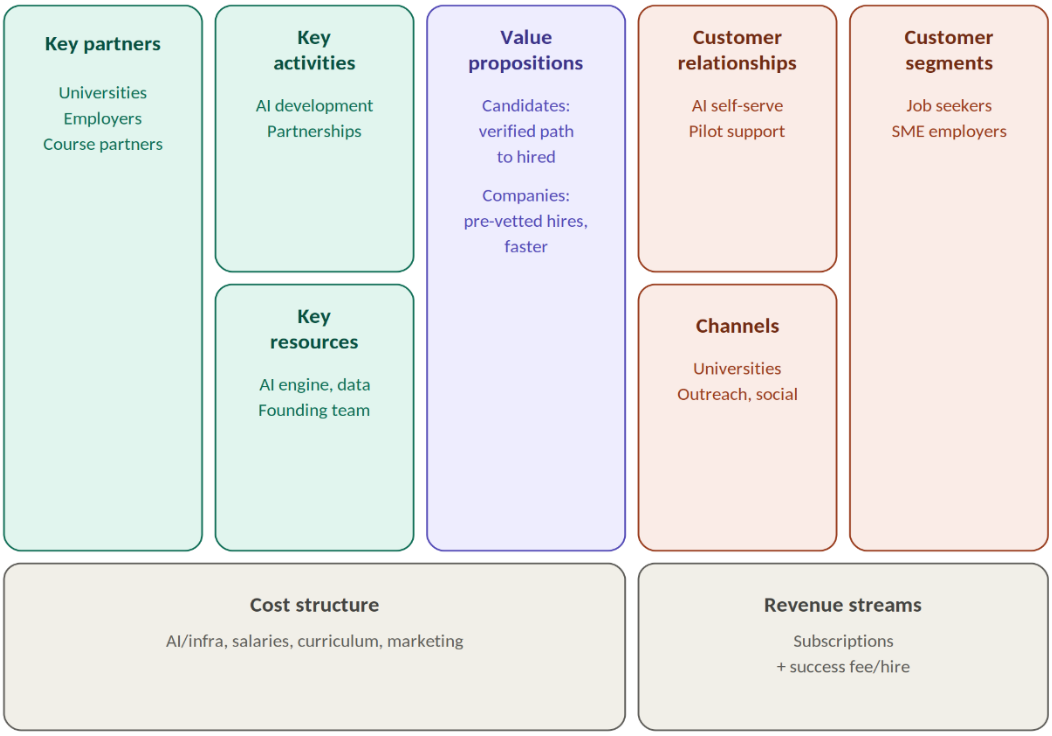
4.3 Initial Go-to-Market Strategy

Skill.IQ's initial go-to-market centers on a pilot program with 10 tech startups and 100 university seniors, designed to test and validate the full learning-to-hire loop end to end. To prove the platform's speed and quality to early company partners, Skill.IQ will offer a "Free First Hire" to each pilot company, removing the cost barrier to trial and generating the platform's first real placement outcomes, which then become the proof points used to convert pilot companies into paying subscribers and to attract the next cohort of candidates and employers.

5. Business Model & Growth Plan

5.1 Business Model Canvas

Skill.IQ — Business Model Canvas



5.2 Partnerships & Funding Strategy

Potential Partners: Universities and career-services offices (for candidate pipeline and pilot access), tech-SME employer associations and startup incubators (for company pilot partners), and third-party course providers such as Coursera or Udemy (for curriculum aggregation rather than content creation).

Funding Strategy: Given Skill.IQ's current idea stage, the funding approach is deliberately staged rather than a single large raise. The first priority is non-dilutive capital resources that don't require giving up equity before there's a working product to value. This includes hackathon and university-innovation prize funding, incubation support through Informatics Institute of Technology's own entrepreneurship programs, and Sri Lanka's newly announced government Startup Fund, which the Ministry of Digital Economy confirmed will launch in early 2026 specifically to support local founders with limited access to venture capital.

In parallel, Axiom V intends to apply to structured, founder-focused programs built for exactly this stage such as the Founder Institute's South Asia cohort and Colombo-based corporate accelerators like John Keells X both of which provide mentorship, structure, and warm investor introductions rather than requiring traction that doesn't yet exist. Only once an MVP is built and pilot results (per Section 4.3) are in hand will Axiom V pursue a small angel or pre-seed round to fund scaling beyond the pilot.

6. Technical Overview & Implementation

6.1 Technical Overview

Skill.IQ's architecture is built around three layers: a conversational AI layer, a media-processing layer for mock interviews, and a standard web application layer connecting candidates and companies.

The conversational AI layer uses a large language model (such as Claude or GPT via API) to power the candidate intake conversation, dynamic learning-path generation, and adaptive follow-up questioning both for candidates describing their goals and for companies describing hiring needs. The media-processing layer handles the voice-based mock interviews: candidate responses are transcribed in real time using a speech-to-text model (such as OpenAI's Whisper or a managed alternative like Deepgram), while a secondary layer analyzes vocal tone, pacing, and confidence markers to support the behavioral/sentiment scoring described in the product overview, alongside LLM-based analysis of the transcript's technical substance. The application layer is a standard web platform which is a React/Next.js frontend serving both the candidate and company dashboards, backed by a Node.js or Python (FastAPI) backend, with PostgreSQL for structured data (users, learning paths, company profiles) and a vector database (such as Pinecone or pgvector) to power skill-based candidate-to-role matching. The entire system is designed to run on managed cloud infrastructure (AWS or GCP) to minimize DevOps overhead at this stage

6.2 Implementation Plan

Development will proceed in four stages aligned with the product roadmap in Section 8.1. The initial build focuses on the conversational intake and learning-path generation engine using off-the-shelf LLM APIs to validate the core experience quickly, without building custom models before the concept is proven. The second stage layers in the mock-interview engine, integrating a speech-to-text API with LLM-based response scoring and starting with text-based technical evaluation before adding voice-tone analysis, to manage complexity for a small early team. The third stage builds the company-facing dashboard and matching logic. Anti-cheating verification (e.g., session integrity checks and response-originality screening) will be scoped once the core interview engine is stable, rather than attempted in parallel with it. Each stage will be tested internally before the Phase 4 pilot described in Section 4.3, so that real user feedback is not assumptions and drives the next round of development priorities.

7. User Experience & Validation

7.1 User Scenario

Meet Aria, a final-year computer science student who has just started thinking seriously about her career but feels unsure where to focus and anxious about interviews. She opens Skill.IQ and starts a conversation with the AI, which asks about her interests, target roles, and current skill level. Based on her answers, the AI builds her a personalized learning path aligned with what employers in her target field actually look for. Over the following weeks, Aria works through the path, completing checkpoints and tests that the platform uses to verify she's genuinely retaining the material, not just clicking through videos.

Once her learning path is complete, the AI schedules her first mock interview. It asks role-specific questions, evaluates both the substance of her answers and her tone, pacing, and confidence, and gives her direct feedback on how to improve. Aria retakes the mock interview a few times, visibly gaining confidence each round, until she passes. She's then assigned a short job-simulation project that mirrors real day-to-day work in her target role, giving her something concrete to speak to in real interviews.

From there, Skill.IQ proactively puts Aria forward to companies looking for someone with her exact profile. A hiring manager at a growth-stage tech company receives a Talent-Drop notification with Aria's Performance Report attached, reaches out directly, and within days Aria has her first real interview, walking in already knowing she's prepared, because the platform has already proven it to her.

7.2 User Feedback *(If Available)*

Include user testing results, testimonials, surveys, or other evidence validating your idea.

8. Roadmap.

8.1 Product Roadmap

Phase 1 (MVP): Build the core conversational AI intake, learning-path generation engine, and progress-checkpoint system for candidates.

Phase 2 (Interview Engine): Develop and test the AI mock-interview system, including behavioral/sentiment analysis and the anti-cheating verification layer.

Phase 3 (Matching Layer): Build the company-facing dashboard, Talent-Drop notification system, and Performance Report generation.

Phase 4 (Pilot & Iterate): Launch the pilot with 10 tech startups and 100 university seniors, refine the matching algorithm based on real hiring outcomes.

8.2 Operational Roadmap

In the near term, the venture is founder-led, with immediate priorities on recruiting a technical co-founder/ML engineer, a full-stack engineer, and a growth/partnerships lead to run the pilot program (as outlined in the pitch deck's "Ask"). As the pilot converts into paying company relationships, the team will scale engineering capacity first, followed by dedicated customer success support for company accounts once dashboard subscriptions launch

8.3 Financial Roadmap

Skill.IQ's near-term financial roadmap is scoped to fund one milestone: a working MVP and completed pilot program, rather than a full company build-out. The estimated funding requirement for this stage is in the range of USD 15,000–30,000 (or the LKR equivalent), allocated primarily toward: bringing on a technical co-founder or contracted developer for MVP build (the largest cost at this stage), AI/API usage costs (LLM calls, speech-to-text processing), cloud hosting, and a small incentive budget to support the "Free First Hire" pilot offer described in Section 4.3. This is intentionally structured as non-dilutive or founder-stage capital, sourced through the channels outlined in Section 5.2 rather than a priced investment round, with a future pre-seed raise considered only once pilot results provide real traction to base a valuation on.

9. Pitch Video

Submit a **1-2 minute pitch video** that clearly convinces the evaluators that your startup idea is worth pursuing.

Your pitch should focus on:

- **The problem** you have identified.
- **Your solution** and how it addresses the problem.
- **Why your solution is important in the present day** and why it deserves to move forward.

The pitch should be delivered in a clear and convincing manner. To ensure fair evaluation, **do not include background music, B-roll footage, motion graphics, or other visual effects**. The focus should be on your explanation and the strength of your idea.

Upload the video to **YouTube as an Unlisted video** and provide the link here.

https://youtu.be/RSr6tH7bqec - Pitch Video

10. Team Vision & Information

Axiom V exists because of a problem I see getting worse, not better: unemployment is rising, and the people hit hardest are graduates who did everything right, studied, earned a degree and still can't find a way into the workforce, often because no one gave them a real chance to prove they were ready. At the same time, companies are burning money on advertising and losing weeks of productivity waiting to fill roles, when the right candidate may already exist and simply never surfaced to them. I'm building Skill.IQ now because this gap is only going to widen as the economy tightens further and I'd rather build the bridge before more people fall into it than after. My purpose with this venture is simple: help people find real opportunity, and help companies find the people who deserve it, without either side paying the current cost of inefficiency and uncertainty.

Provide details of all team members.

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